

|             |           |             |                            |                 |   |                       |   |   |   |   |
|-------------|-----------|-------------|----------------------------|-----------------|---|-----------------------|---|---|---|---|
| Course Code | 21CSE354T | Course Name | FULL STACK WEB DEVELOPMENT | Course Category | E | PROFESSIONAL ELECTIVE | L | T | P | C |
|             |           |             |                            |                 |   |                       | 2 | 1 | 0 | 3 |

|                            |                     |                               |     |                     |     |
|----------------------------|---------------------|-------------------------------|-----|---------------------|-----|
| Pre-requisite Courses      | Nil                 | Co-requisite Courses          | Nil | Progressive Courses | Nil |
| Course Offering Department | School of Computing | Data Book / Codes / Standards |     |                     | Nil |

| Course Learning Rationale (CLR): | The purpose of learning this course is to:                     | Program Outcomes (PO) |                  |                                 |                                            |                   |                          |                              |        |                        |               |                        |                    | Program Specific Outcomes |       |       |
|----------------------------------|----------------------------------------------------------------|-----------------------|------------------|---------------------------------|--------------------------------------------|-------------------|--------------------------|------------------------------|--------|------------------------|---------------|------------------------|--------------------|---------------------------|-------|-------|
| CLR-1:                           | introduce the Web Fundamentals                                 | 1                     | 2                | 3                               | 4                                          | 5                 | 6                        | 7                            | 8      | 9                      | 10            | 11                     | 12                 |                           |       |       |
| CLR-2:                           | introduce the Client-side scripting with react.js              | Engineering Knowledge | Problem Analysis | Design/development of solutions | Conduct investigations of complex problems | Modern Tool Usage | The engineer and society | Environment & Sustainability | Ethics | Individual & Team Work | Communication | Project Mgt. & Finance | Life Long Learning | PSO-1                     | PSO-2 | PSO-3 |
| CLR-3:                           | introduce the Database Connectivity                            |                       |                  |                                 |                                            |                   |                          |                              |        |                        |               |                        |                    |                           |       |       |
| CLR-4:                           | introduce the Spring Framework with Basic Concepts             |                       |                  |                                 |                                            |                   |                          |                              |        |                        |               |                        |                    |                           |       |       |
| CLR-5:                           | introduce the Spring Boot and Micro-Services                   |                       |                  |                                 |                                            |                   |                          |                              |        |                        |               |                        |                    |                           |       |       |
| Course Outcomes (CO):            | At the end of this course, learners will be able to:           |                       |                  |                                 |                                            |                   |                          |                              |        |                        |               |                        |                    |                           |       |       |
| CO-1:                            | understand the Static Web Page Application                     | 3                     | 2                | 2                               | -                                          | -                 | -                        | -                            | -      | -                      | -             | -                      | -                  | 2                         | -     | -     |
| CO-2:                            | apply scripting at client side                                 | 3                     | 2                | 2                               | -                                          | -                 | -                        | -                            | -      | -                      | -             | -                      | -                  | 2                         | -     | -     |
| CO-3:                            | connect with Database to do CRUD operations                    | 3                     | 2                | 2                               | -                                          | -                 | -                        | -                            | -      | -                      | -             | -                      | -                  | 2                         | -     | -     |
| CO-4:                            | develop, Maintain and applications using Spring Boot Framework | 3                     | 2                | 3                               | -                                          | -                 | -                        | -                            | -      | -                      | -             | -                      | -                  | 2                         | -     | -     |
| CO-5:                            | ability to use Microservices                                   | 3                     | 2                | 2                               | -                                          | -                 | -                        | -                            | -      | -                      | -             | -                      | -                  | 2                         | -     | -     |

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| <b>Unit-1 - HTML, CSS Overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>9 Hour</b> |
| HTML Overview: Structuring an HTML Document, Validating and debugging your code, Working with Fonts, Text Blocks, Lists and Tables, Using External and Internal Links, Working with Colors, Images, and Multimedia. CSS Overview: Understanding Cascading Style Sheets, Working with Margins, Padding, Alignment and Floating, Understanding the CSS Box Model and Positioning, Using CSS to do more with Lists, Text, and Navigation, Creating Layouts Using modern CSS Techniques, Taking Control of Backgrounds and Borders, Using CSS Transformations and Transitions, Animating with CSS and the Canvas.                                                                                                                              |               |
| <b>Unit-2 - JavaScript, React JS Overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>9 Hour</b> |
| JavaScript Overview: Understanding JavaScript, Understanding Dynamic Websites and HTML5 Applications, Getting Started with JavaScript Programming, working with the Document Object Model (DOM), Using JavaScript Variables, Strings, and Arrays, Controlling Flow with Conditions and Loops, Responding to Events and Using Windows, JavaScript Best Practices., Using Third-Party JavaScript Libraries and Frameworks. React JS: The Foundation of React, JSX, All About Components of React, Events, Forms, Refs, Styling React, Routing, Hooks: Rules of Hooks, The Built-in Hooks (useState, useEffect, useContext, useMemo, useRef), Writing Custom Hooks, Labeling Custom Hooks with useDebugValue, Finding and Using Custom Hooks. |               |
| <b>Unit-3 - SQL, JDBC Overview</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>9 Hour</b> |
| SQL Overview: Basics of SQL: Data basics, Retrieval: Basic Selection, Joins, Set Queries: UNION, INTERSECT, and EXCEPT, Subqueries, Modifying Data: Insert, Delete, Update, Creating, Deleting, and Altering Tables, JDBC: Introduction to JDBC: Setting up the database, connecting to a Database: The Connection Interface, connecting to the database using Driver Manager, Querying and Updating the Database: Statement Interface, Result Set Interface, Querying and Updating the Database.                                                                                                                                                                                                                                          |               |
| <b>Unit-4 - Spring Boot Framework-Part-1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>9 Hour</b> |
| Spring Boot Framework-Part-1: Basic concepts: Spring, Spring Boot, Testing Basics, Testing in Spring Boot. A Basic Spring Boot Application: Setting up the Development Environment, The Skeleton Web App, and Spring Boot Auto configuration, Three-Tier, Three-Layer Architecture, Modeling our Domain, Business Logic, and Presentation Layer.                                                                                                                                                                                                                                                                                                                                                                                           |               |

**Unit-5 - Spring Boot Framework-Part-2****9 Hour**

Spring Boot Framework-Part-2: The Data Layer: The Data Model, choosing a Database, Spring Boot Data JPA, Entities, Repositories, Storing Users and Attempts. Microservices: Moving to Microservices, Architecture Overview, Designing and Implementing the New Service, User Interface UI

|                           |                                                                                                                                                                                                                                   |                                                                                                                                                                                                        |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Learning Resources</b> | 1. Sams Teach Yourself HTML, CSS, and JavaScript All in One, Julie Meloni, First Edition 2019, Pearson. (Unit - I, Unit - II)                                                                                                     | 6. Programming the World Wide Web, by Robert W. Sebesta, Eighth Edition - 2014, Published by Pearson.                                                                                                  |
|                           | 2. BEGINNING ReactJS Foundations Building User Interfaces with ReactJS: AN APPROACHABLE GUIDE, Chris Minnick, First Edition 2022, Published by John Wiley & Sons. (Unit - II)                                                     | 7. Internet and World Wide Web How to Program, Paul Deitel, Harvey Deitel, and Abbey Deitel, Fifth Edition - 2011, Published by Prentice Hall.                                                         |
|                           | 3. SQL Practical Guide for Developers, Michael J. Donahoo & Gregory D. Speegle, 2005, Morgan Kaufmann Publishers an Imprint of Elsevier. (Unit - III)                                                                             | 8. Database Programming with JDBC and Java, by George Reese, Second Edition - 2000, Published by O'Reilly Media.                                                                                       |
|                           | 4. Oracle Certified Professional JavaSE 8 Programmer, Exam 1Z0-809 A Comprehensive OCPJP 8 Certification Guide, by S G Ganesh, Hari Kiran & Tushar Sharma, 2016, APRESS publisher. (Unit - III)                                   | 9. Expert Oracle JDBC Programming, by R. M. Menon, First Edition - 2005, Published by Apress.                                                                                                          |
|                           | 5. Learn Microservices with Spring Boot: A Practical Approach to RESTful Services Using an Event-Driven Architecture, Cloud-Native Patterns, and Containerization, Moisés Macero García, Second Edition - 2020, APRESS Publisher. | 10. Pro Spring MVC with WebFlux: Web Development in Spring Framework 5 and Spring Boot 2, by Marten Deinum and Iuliana Cosmina, Second Edition - 2022, Published by Apress.                            |
|                           |                                                                                                                                                                                                                                   | 11. Full stack development with Spring Boot and React: build modern and scalable full stack applications using the power of Spring Boot and React, by Juha Hinkula, Third Edition - 2022, Published by |
|                           |                                                                                                                                                                                                                                   | 12. Spring Boot in Practice, Somnath Musib, First Edition - 2022, Manning Publications.                                                                                                                |

| Learning Assessment |                              |                                                  |          |                                      |          |                                                   |          |
|---------------------|------------------------------|--------------------------------------------------|----------|--------------------------------------|----------|---------------------------------------------------|----------|
|                     | Bloom's<br>Level of Thinking | Continuous Learning Assessment (CLA)             |          |                                      |          | Summative<br>Final Examination<br>(40% weightage) |          |
|                     |                              | Formative<br>CLA-1 Average of unit test<br>(50%) |          | Life-Long Learning<br>CLA-2<br>(10%) |          |                                                   |          |
|                     |                              | Theory                                           | Practice | Theory                               | Practice | Theory                                            | Practice |
| Level 1             | Remember                     | 15%                                              | -        | 15%                                  | -        | 15%                                               | -        |
| Level 2             | Understand                   | 25%                                              | -        | 20%                                  | -        | 25%                                               | -        |
| Level 3             | Apply                        | 30%                                              | -        | 25%                                  | -        | 30%                                               | -        |
| Level 4             | Analyze                      | 30%                                              | -        | 25%                                  | -        | 30%                                               | -        |
| Level 5             | Evaluate                     | -                                                | -        | 10%                                  | -        | -                                                 | -        |
| Level 6             | Create                       | -                                                | -        | 5%                                   | -        | -                                                 | -        |
|                     | Total                        | 100 %                                            |          | 100 %                                |          | 100 %                                             |          |

| <b>Course Designers</b> |                                            |                            |
|-------------------------|--------------------------------------------|----------------------------|
| Experts from Industry   | Experts from Higher Technical Institutions | Internal Experts           |
| Dr.Mariappan, Uber      |                                            | 1. Mr.Suresh Anand, SRMIST |
|                         |                                            | 2. Ms. N.Anbarasi, SRMIST  |